

Differential Location Analysis of Transcription Factor

A novel approach to detecting cellular responses to environmental changes

Progress Report II by Junguk Hur

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Introduction

This project is to reveal underlying correlation among different transcription factors in a quantitative way by implementing differential location analysis of ChIP-on-Chip combined with other datasets like Microarray, Protein-Protein Interaction, and Motif analysis. I also aim at characterizing primary and secondary responses to environmental changes which can not be revealed by Microarray experiments alone.

With the advent of Chromatin Immunoprecipitation on microarray, or so called ChIP-on-Chip, along with the availability of complete genome sequences, genome-wide location analysis has become one of the hot areas in genomics researches. Based on the currently available 203 yeast transcription factor ChIP-on-Chip data under various conditions from Richard Young group (<http://web.wi.mit.edu/young/>) at MIT, we want to exploit the hidden mechanisms of transcriptional regulation and cooperation among different TFs in quantitative ways with the integration of other high-throughput Microarray gene expression, protein-protein interaction and sequence data.

Project components

This project involves in various areas of bioinformatics approaches such as high-throughput data processing, integration, statistical analyses, and sequence based motif analyses. The following summarizes what I have done so far in these aspects. (Details can be found either at the project proposal or at the first progress report)

1. Primary Data Handling
 - a. Data acquisition
 - i. ChIP-on-Chip data: 203 yeast transcription factors in 12 different conditions from the Young Lab at MIT
 - ii. Gene expression data
 1. cDNA microarray for similar condition of heat-shock and oxidative stress (H_2O_2) to ChIP-on-Chip experiments.
 2. Oligonucleotide microarray for oxidative stress (H_2O_2)
 3. Yeast transcription factors in TRANSFAC database.
 - b. Data pre-processing
 - i. ChIP-on-Chip data (ratio and p-value) was pre-processed by removing null data and combining p-value and ratio data.
2. Differential analysis of Chip-on-Chip
 - a. We devised a correlation score (p) of ratio between different conditions for every available pairwise condition for different p-value thresholds.

3. Statistical Analysis
 - a. In order to confirm the distribution of the highly differentially bound genes is not following normal distribution and rather skewed, Chi-Square tests have been performed against normal distribution.
4. Categorization of three differential binding patterns of TFs
 - a. 154 pairs of conditions have been categorized into three different groups depending on their differential binding patterns
 - i. UP group: TFs bind more strongly to most of the target regions in the second condition. (This implies increased binding affinity)
 - ii. DOWN group: TFs bind less strongly to most of the target regions in the second condition. (This implies decreased binding affinity)
 - iii. BOTH group: The number of bound regions are about the same in UP and DOWN portion.
5. Sequence Analysis
 - a. Each promoter regions were scanned for possible TFBS via MATCH program against TRANSFAC databases.

Result

Table 1 shows skewness of the distribution of differentially binding TFs in H2O2 (oxidative stress status) from ChIP-on-Chip data and microarray data. Only low concentration of H2O2 (0.4mM) from ChIP-on-Chip data was considered. Out of 27 transcription factors, 22 have shown skewness from normal distribution with very low p-value which implies significant skewed distribution far from random distribution.

TF	Binding Assay				Log Ratio (Microarray)		
	H2O2Hi		H2O2Lo		H2O2		
	4mM (30min)		0.4mM (20min)		cDNA		
	3B		27		0.3mM_20Min	0.32mM_20Min	0.32mM_20Min
YAP6	6.80E-03	SKEW	8.80E-05	SKEW	0.446	0.862	--
CIN5	7.80E-06	SKEW	1.10E-09	SKEW	1.039	2.441	5.42
ROX1	2.00E-06	SKEW	4.20E-02	SKEW	-0.07	1.152	2.7
RPN4			1.60E-11	SKEW	0.576		1.95
YAP7	2.20E-16	SKEW	2.20E-16	SKEW	-0.596	0.902	1.89
HAP4			4.50E-07	SKEW	-0.485	-0.207	0.88
REB1	1.20E-03	SKEW	2.90E-12	SKEW	-0.245	-0.695	0.24
YAP1	4.20E-13	SKEW	2.10E-03	SKEW		2.134	0.24
XBP1			3.30E-02	SKEW		1.057	0.18
MSN4	3.30E-04	SKEW	9.20E-05	SKEW	0.565	0.903	0.15
PHO2	2.50E-04	SKEW	1.40E-06	SKEW	0.289	-0.361	0.08
PUT3			3.90E-04	SKEW	0.018	0.866	-0.18
MSN2	2.20E-16	SKEW	8.00E-10	SKEW		0.287	-0.33
RIM101	9.30E-02	NOR	5.10E-03	SKEW	-0.118		-0.68
SFP1	3.30E-01	NOR	6.60E-03	SKEW	-0.382	0.139	-0.71
HSF1	2.20E-16	SKEW	3.30E-11	SKEW	0.277		-0.82
MBP1	1.10E-08	SKEW	2.90E-08	SKEW	-0.035	-0.055	-1.19
FKH2	3.50E-06	SKEW	7.80E-11	SKEW	-1.319	0.158	-1.34
RCS1	7.40E-02	NOR	7.60E-07	SKEW	-0.581	0.571	-1.36
SKN7	2.40E-06	SKEW	9.60E-11	SKEW			-1.78
PDR1			1.30E-09	SKEW	0.107	1.361	-2.24
AFT2	1.20E-02	SKEW	7.40E-10	SKEW		1.574	
MOT3	4.50E-02	SKEW	1.30E-01	NDR	-0.765	0.022	--
RTG3	2.20E-06	SKEW	6.90E-02	NDR	0.244		--
YIL206C	6.50E-03	SKEW	1.90E-01	NDR			-0.41
RPH1	1.40E-01	NOR	6.10E-02	NDR	-0.006	-0.604	-0.88
NRG1	8.60E-03	SKEW	2.20E-01	NDR			

Table 1 Comparison between Binding Assay and Microarray

Table 2 summarizes the comparison between ChIP-on-Chip and Microarray data analysis. About 60% percent of TFs do not show significant expressional changes in the Microarray data.

ChIP-Chip	H2O2 Lo (27 TFs)					
	Previous		After		Both	Any
Microarray	Match	Not	Match	Not	Match	Match
	24	3	22	5	14	26
Skewed	21	1	20	2	13 93%	22 85%
Fold change	>=2	7	.	9	.	4 31%
	0 ~ 2	14	.	11	.	9 69%
Normal	3	2	2	3	1 7%	4
Fold change	>=2	0	.	0	.	.
	0~2	3	.	2	.	1 4

Table 2 Summary of ChIP-on-Chip and Microarray comparison

We have categorized TFs into three different groups depending on the differential binding patterns. The UP group has 28 TF/condition pairs while the BOTH group and the DOWN group have 43 and 34 pairs respectively. Full list of these groups are represented in the Table 3. Examples of the each group are illustrated in the Figure 1.

CLASS	UP		BOTH			DOWN		
Ratio	Ra > 0.67		0 < Ra or Rb < 0.67			Rb > 0.76		
TF & Condition	DIG1_Alpha	MBP1_H2O2Lo	GAT1_HEAT	DAL81_SM	YAP1_H2O2Lo	CBF1_SM	AFT2_H2O2Lo	CHA4_SM
	DIG1_BUT14	HAP4_SM	RPN4_H2O2Lo	PHO4_Pi	GCR2_SM	GCN4_SM	MSN2_H2O2Lo	UME6_H2O2Hi
	MSN2_HEAT	YAP3_H2O2Hi	FHL1_H2O2Hi	GZF3_H2O2Hi	SFP1_H2O2Hi	GCN4_RAPA	CIN5_H2O2Lo	SIG1_H2O2Hi
	PHD1_BUT14	PHO2_Pi	HAP4_H2O2Lo	YAP6_H2O2Lo	YAP6_H2O2Hi	GLN3_SM	RAP1_SM	RDS1_H2O2Hi
	PHD1_BUT90	ARR1_H2O2Hi	MSN4_Acid	MSN4_H2O2Lo	NRG1_H2O2Hi	HSF1_H2O2Hi	FHL1_RAPA	MSS11_BUT14
	RLM1_BUT14	CAD1_SM	MBP1_H2O2Hi	HAP5_SM	RTG1_SM	MSN2_H2O2Hi	THI2_Thi	GZF3_RAPA
	STE12_BUT90	GAL4_RAFF	RCS1_H2O2Hi	PHO2_SM	ARO80_SM	STE12_Alpha	GLN3_RAPA	
	DIG1_BUT90	RPH1_H2O2Lo	PHO2_H2O2Hi	MSN4_H2O2Hi	AFT2_H2O2Hi	YAP7_H2O2Hi	ASH1_BUT14	
	YAP5_H2O2Hi	MET28_SM	ROX1_H2O2Hi	PUT3_SM	STE12_BUT14	YAP7_H2O2Lo	DAL82_RAPA	
	YAP1_H2O2Hi	BAS1_SM	RTG3_SM	MET32_SM	SOK2_BUT14	REB1_H2O2Lo	SKN7_H2O2Hi	
	YAP1_HEAT	RIM101_H2O2Lo	IME4_H2O2Hi	HAP2_RAPA	XBP1_H2O2Lo	HSF1_H2O2Lo	FKH2_H2O2Hi	
	FKH2_H2O2Lo	YJL206C_H2O2Hi	CIN5_H2O2Hi	ARG80_SM	MIG2_H2O2Hi	SKN7_H2O2Lo	MSN2_RAPA	
	MGA1_H2O2Hi	MAC1_H2O2Hi	ADR1_HEAT	REB1_H2O2Hi	ROX1_H2O2Lo	GAT1_RAPA	SFP1_H2O2Lo	
	PDR1_H2O2Lo	KSS1_BUT90	RTG3_H2O2Hi	UME1_H2O2Hi	MAL33_H2O2Hi	DAL82_SM	STP1_SM	
				MOT3_SM				
Num	28		43			34		

Table 3 Categorization of TFs depending on the differential binding patterns

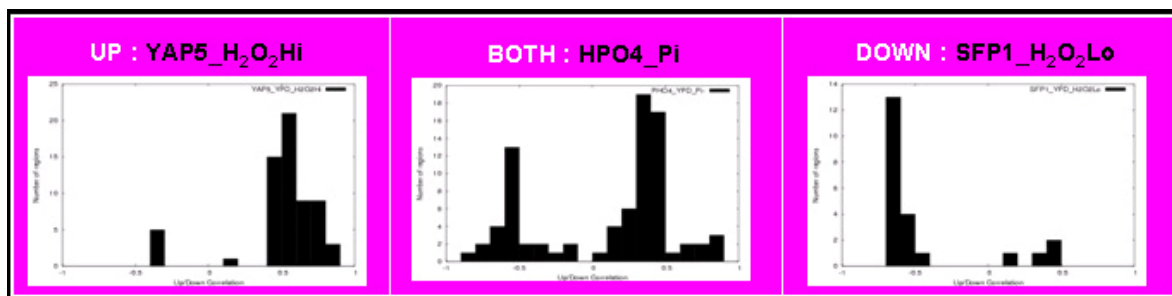


Figure 1 Examples of categorized TF groups

Discussion

From the comparative analysis of genome-wide binding assay and gene expression assay, we have found that most of the transcription factors do not change their transcriptional level (mRNA) in response to environmental changes. This has been shown to be statistically meaningful by the correlation analysis and chi-square distribution test.

I am currently working on sequence analysis to identify putative binding sites of other transcription factors that might affect the binding affinity of given transcription factors. The BOTH group has been chosen as primary research target since they have both up-differentially binding and down-differentially binding regions at the same time. We are expecting sequence signatures or profiles that can distinguish these two contracting patterns by specifically analyzing the sequences of the putative binding sites.

Future works include the integration of proteomic data (post translational modification) or protein-protein interaction data to check for any possible effects of this biochemical interaction or modification on the differential transcription binding. But this work is beyond my Capstone project scope.

Progress Bar

Sep	Oct	Nov	Dec	'06 Jan	Feb	Mar	Apr
Data Processing		Statistical analysis		Categorization	TFBS Analysis	Web page construction	Paper submission
						75% overall (90% analysis)	

Note

This project was presented at the CGB Roundtable (<http://cgb.indiana.edu>) on Jan. 26, 2006. The presentation file is available at my homepage (<http://darwin.informatics.indiana.edu/juhur>) and it contains selected examples of current project results. All other materials are not to be disclosed yet.